

Module 07: Communication in Robotics — Networked Control Systems

Learning Objectives

1. Define **networked control** as control-estimation implemented over communication networks with delays, packet loss, and bandwidth constraints.
2. Analyze how **communication delays, quantization, and network topology** affect state estimation and control performance.
3. Apply Active Inference precision concepts to handle communication imperfections in multi-robot systems.

Introduction

Modern robotic systems increasingly operate over communication networks — sensor data is transmitted wirelessly, control commands cross network boundaries, and multiple robots coordinate through shared communication channels. Network imperfections (delay, jitter, packet loss, bandwidth limits) directly impact the quality of the Active Inference loop.

Key Concepts

1. Communication Delays and State Estimation

Network delays mean observations arrive stale — the sensor reading was taken Δt seconds ago:

- The estimator must account for this delay by "predicting forward" from the observation's timestamp to the current time using the B matrix
- Larger delays reduce the effective precision of observations — the forward-predicted uncertainty grows with delay length
- Active Inference handles this naturally: delayed observations are treated as low-precision observations, automatically receiving less weight in the state estimate

2. Packet Loss and Robust Estimation

When observations are lost (dropped packets), the estimator must operate in open-loop — predicting without observation correction:

- **Short outages:** The B matrix carries the prediction forward with growing uncertainty; when observations resume, the correction snaps the estimate back
- **Long outages:** Prediction uncertainty grows until the state estimate becomes unreliable; the controller should shift to conservative behavior (slow down, maintain position)
- **Active Inference** handles packet loss by reducing observation precision to zero during outages — the estimator relies entirely on its prior predictions

3. Bandwidth-Constrained Communication

Limited bandwidth forces robots to compress their communications:

- **Event-triggered communication:** Only transmit when the prediction error exceeds a threshold — reducing bandwidth by 60-90% compared to periodic transmission with minimal performance degradation
- **Quantized observations:** Sensor readings are discretized to reduce transmission size — introducing quantization noise that appears as reduced observation precision
- **Compressed belief sharing:** Rather than sharing full probability distributions, agents share sufficient statistics (mean and variance, or innovation sequences)

4. Network Topology and Information Flow

The network topology determines how quickly information propagates through a multi-robot system:

- **Star topology** (all robots communicate with a central node): Fast consensus but single point of failure
- **Mesh topology** (robots communicate with neighbors): Robust but slower information propagation
- **Hierarchical topology** (team leaders relay to a coordinator): Balances speed and robustness

Each topology implements a different message-passing architecture for distributed Active Inference.

Applications

- **Teleoperated surgical robot:** A surgeon operates a remote surgical robot over a network with 50-200ms round-trip delay. The system uses predictive display (showing the surgeon a simulated view of the predicted robot state rather than the delayed actual state) and wave variable control (guaranteeing stability despite delay) — both are Active Inference solutions to the delay problem.
- **Planetary rover team:** A team of Mars rovers coordinates through a low-bandwidth, high-delay communication link — sharing compressed occupancy maps and task allocation messages using event-triggered protocols that minimize uplink usage.

Conclusion

Networked control systems implement Active Inference over imperfect communication channels. Delays, packet loss, bandwidth limits, and network topology all affect the quality of distributed inference. Understanding these effects through precision and prediction error enables robust multi-robot system design.